FUTURE STATE JOURNEY MAP

# Batch Retrieval for Equipment Review Workflows

Part Identification Repository · Pilot
All research data is simulated

**PERSONA**
Equipment review engineer, an engineer who verifies parts for equipment reviews and is under delivery pressure. The engineer starts the workflow, which may be built in collaboration with others.

**GOAL**
Complete a BOM-wide equipment review on schedule without manually looking up each part.

## JOURNEY TIMELINE

| STAGE | USER ACTION | INTERNAL STATE | PAIN POINT ADDRESSED |
|---|---|---|---|
| 1 **Submit** | Start the workflow, which submits 1 to 1,000 part numbers in one authenticated request<br>→ **No sequential loops** | Confident the sanctioned path works<br>→ **No urge to improvise a workaround** | Replaces custom wrappers<br>→ **No hundreds of sequential GETs** |
| 2 **Retrieve** | Receive every valid part, with each invalid part flagged, in under 1,500 ms (target)<br>→ **Review starts immediately** | Trusts the system is fast<br>→ **Stays focused on engineering judgment** | One batch call avoids rate limits<br>→ **No halfway failures**<br>Invalid part numbers are flagged without failing the batch<br>→ **No full restart after a mid-run failure** |
| 3 **Analyze** | Pass lean, UI-free payload to AI agent<br>→ **≥75% fewer tokens** (benefit) | Empowered, as AI cross-references the BOM<br>→ **Effort shifts to decisions** | Strips layout, image URLs, styling<br>→ **Ends payload bloat and token waste** |
| 4 **Deliver** | Finish BOM-wide equipment review on schedule<br>→ **Deadline met without workarounds** | Secure and compliant<br>→ **No insecure CSV dumps to explain** | Solves the single-lookup limit<br>→ **Removes the need for scraping data from the legacy UI and CSV dumps**<br>Security visibility assumed through existing activity logging and reporting |

## COMPETITIVE ADVANTAGES OVER THE SHADOW WORKAROUND

**Speed**
1 to 1,000 parts in one request, returned in under 1,500 ms (target)
→ **Replaces hundreds of sequential requests**

**Security**
Sanctioned path
→ **Removes the need for shadow workarounds**
Visibility assumed through existing activity logging and reporting

**Cost**
Stripped, AI-ready payloads
→ **≥75% lower AI computing cost** (benefit)

**Known risk:** The 1,500 ms target for 1,000 parts depends on backend query efficiency. N+1 query degradation (Critical, see Thematic Synthesis) is a known risk to resolve for the pilot and before general production release.

Assumptions: All research data is simulated. The feature is piloted with a select team of engineers vetted for security. Role-based access control, activity logging and reporting are assumed, not built. Full security guardrails are required before general production release.